

Multiple Choice (10 marks)

1. Which of the following has the greatest potential for compromising a user's personal privacy?
 - (a) A group of cookies stored by the user's Web browser
 - (b) The Internet Protocol (IP) address of the user's computer
 - (c) The user's e-mail address
 - (d) The user's public key used for encryption
2. In Python 3, let $r = \text{lambda } q: q * 2$. What is $r(3)$?
 - (a) 2
 - (b) 6
 - (c) 3
 - (d) 1
3. Are Python variable names case-sensitive?
 - (a) Yes
 - (b) No
 - (c) It's machine-dependent
 - (d) None of the above
4. In Python 3, what is the output of `print list[1:3]` if `list = [ 'abcd', 786 , 2.23, 'john', 70.2 ]`?
`786 , 2.23, 'john', 70.2`
 - (a) `abcd`
 - (b) `786, 2.23`
 - (c) `None of the above.`
5. Let $x = 8$. What is $x >> 1$ in Python 3?
 - (a) 3
 - (b) 4
 - (c) 2
 - (d) 8

True / False (10 marks)

Answer True or False

6. True or False: The output of the expression `4*1**4` in Python is 1.
7. True or False: The condition `a < b` is sufficient to ensure the expression evaluates to true for integers `a`, `b`, and `c`.
8. True or False: Python supports automatic garbage collection, which helps manage memory efficiently.
9. True or False: In Python 3, accessing the last character of the string `'abc'` with an index of -1 results in an error.
10. True or False: In Python 3, the expression `b[::2]` retrieves the elements at even indices, resulting in `[11, 15, 19]`.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the selection sort algorithm, focusing on the number of comparisons it makes. Why is this number unaffected by the initial arrangement of elements? Provide examples to support your explanation.
12. Discuss the characteristics of phishing attacks and analyze which types of communications are least likely to be indicative of such attacks, providing examples to support your reasoning.

End of Paper